Quantitative Readout Accountability for Finite Centers

Live review note of June 20, 2026

Abstract

The exact Team 1 support/readout criterion says that finite central rank is recoverable only when the supplied readouts are faithful on the physical finite-center contrast space, after null/gauge/inadmissible directions have been quotiented. This note records the finite-precision strengthening. If

$$R : V_F = \ell^\infty(F)/\mathbb{C}1 \longrightarrow W$$

is the declared central readout map, then exact accountability is $\ker R = 0$. Noisy accountability additionally requires a lower inverse modulus

$$\alpha(R) = \inf_{\|v\|=1} \|Rv\| > 0$$

large enough for the error budget. If two contrasts are compatible with the same data ball of radius ε, then their distance is at most $2\varepsilon/\alpha(R)$. Thus a formally separating but badly conditioned readout is not a referee-grade finite-rank or binary-rank measurement unless its separation margin dominates the stated uncertainty. This is standard finite-dimensional inverse stability; the possible value is the explicit black-hole/AQFT finite-center warning.

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1 Primary sources

The boundary below was checked against Blackwell's comparison of experiments [1, 2], Le Cam's sufficiency and approximate sufficiency [3], Petz sufficiency for von Neumann algebras [4], and Buscemi's quantum statistical comparison theorem [5]. These are positive sufficiency/deficiency frameworks. No novelty is claimed for the linear stability estimate.

2 Exact and noisy readout

Let F be finite and set

$$V_F = \ell^\infty(F)/\mathbb{C}1.$$

Let $R : V_F \to W$ be the finite-dimensional readout map generated by the declared central probes, effects, sector records, recovery task, spectroscopy channel, or other finite-center observation. Fix norms on V_F and W. Define

$$\alpha(R) = \inf_{\|v\|=1} \|Rv\|.$$

Proposition 1 (exact accountability)**.** *The readout R separates all finite central contrasts if and only if $\alpha(R) > 0$. In finite dimension this is equivalent to* $\ker R = 0$*, and hence to* $\operatorname{rank} R = \dim V_F$*.*

Proof. If $\alpha(R) > 0$, then $Rv = 0$ implies $\alpha(R)\|v\| \le \|Rv\| = 0$, so $v = 0$. Conversely, if $\ker R = 0$, the continuous function $v \mapsto \|Rv\|$ has a positive minimum on the compact unit sphere of V_F. □

Proposition 2 (noisy accountability)**.** *Assume $\alpha(R) > 0$. If two contrasts $v, v' \in V_F$ are compatible with the same observed readout $y \in W$ within error ε,*

$$\|Rv - y\| \le \varepsilon, \qquad \|Rv' - y\| \le \varepsilon,$$

then

$$\|v - v'\| \le \frac{2\varepsilon}{\alpha(R)}.$$

Consequently contrasts separated by distance at least δ are distinguishable at error ε only if

$$\alpha(R)\delta > 2\varepsilon.$$

Proof. The triangle inequality gives

$$\|R(v - v')\| \le \|Rv - y\| + \|Rv' - y\| \le 2\varepsilon.$$

By definition of $\alpha(R)$,

$$\alpha(R)\|v - v'\| \le \|R(v - v')\|,$$

which proves the estimate. The distinguishability condition is the same inequality read contrapositively for a required contrast resolution δ. □

Proposition 3 (null or gauge quotient)**.** *Let $N \subset V_F$ be a declared null, gauge, or inadmissible subspace and let $\bar{R} : V_F/N \to W$ be the induced readout. Propositions 1 and 2 hold with V_F replaced by V_F/N and $\alpha(R)$ replaced by $\alpha(\bar{R})$.*

Proof. Apply the previous two propositions to the finite-dimensional quotient space. □

3 Binary-rank consequence

A noisy operational claim of finite central rank, and especially a claim of binary rank $q = 2$, requires more than a formally separating family of questions. It requires:

Premise	Role
Physical contrast space V_F/N	States which finite central directions are physical rather than null, gauge, or inadmissible.
Full exact rank	Ensures no physical finite central contrast is in the kernel.
Lower inverse modulus $\alpha(\bar{R})$	Converts formal separation into a stable error-budget statement.
Resolution inequality $\alpha(\bar{R})\delta > 2\varepsilon$	Ensures the claimed rank-breaking contrast is larger than uncertainty.
Maximality, finite-alphabet law, or direct spectroscopy	Prevents a stable binary readout from being merely a binary quotient of a larger hidden center.

Thus exact accountability answers the algebraic question, while quantitative accountability answers the referee's finite-precision question.

4 Referee question

> In the target black-hole/AQFT reconstruction setting, what is the lower readout modulus or equivalent Le Cam/deficiency/error-budget statement that makes the finite central rank claim stable?

If no such modulus or deficiency bound is supplied, Team 1 should treat the rank or $q = 2$ statement as conditional. If the bound is supplied, the finite-center obstruction has an explicit positive exit at the stated precision.

References

[1] D. Blackwell, *Comparison of Experiments*, Proceedings of the Second Berkeley Symposium on Mathematical Statistics and Probability, 1951, pp. 93–102, Project Euclid.

[2] D. Blackwell, *Equivalent Comparisons of Experiments*, Ann. Math. Statist. 24 (1953), 265–272.

[3] L. Le Cam, *Sufficiency and Approximate Sufficiency*, Ann. Math. Statist. 35 (1964), 1419–1455.

[4] D. Petz, *Sufficient Subalgebras and the Relative Entropy of States of a von Neumann Algebra*, Commun. Math. Phys. 105 (1986), 123–131, Project Euclid PDF.

[5] F. Buscemi, *Comparison of Quantum Statistical Models: Equivalent Conditions for Sufficiency*, Commun. Math. Phys. 310 (2012), 625–647, arXiv:1004.3794.